

Puzzle sheet

Scientific notation

'Money has never made a man happy yet, nor will it. There is nothing in its nature to produce happiness.'

16	13	18	19	14	2	18	15	19	15	17	13	15	9					
16	13	18	19	14	2	18	13	18	6	15	17	16	9					
8	17	9	16	18	15	4	14	7	8	16	9	7	8	12	12	8	17	5
15	11	15	1	10	10	19	8	16	19	15	3	18	9	14	17	18		

The numbers in the grid above match the question numbers. The answer to each question below has a corresponding letter as shown by the 'Key' on the next page. Answer the questions and then match the answers to letters in the 'Key'. Fill in the boxes above with the letters that match the questions, to discover a quote from American inventor Benjamin Franklin.

Questions

Write each of the following in scientific notation.

- | | | | |
|-----------------------|-------|-------------------|-------|
| 1 8 390 000 | _____ | 2 0.000 000 421 6 | _____ |
| 3 70 050 | _____ | 4 0.0099 | _____ |
| 5 0.000 000 000 037 3 | _____ | 6 480 200 | _____ |

Write each of the following as an ordinary number.

- | | | | |
|---------------------------|-------|---------------------------|-------|
| 7 1.54×10^7 | _____ | 8 3.2×10^{-2} | _____ |
| 9 7.908×10^5 | _____ | 10 6.005×10^{12} | _____ |
| 11 4.606×10^{-4} | _____ | 12 5.07×10^{-7} | _____ |

Calculate the following.

- | | |
|--|-------|
| 13 $9 \times 10^{12} \div (3 \times 10^5)$ | _____ |
| 14 $7 \times 10^{-4} \times 5.1 \times 10^{-9}$ | _____ |
| 15 $(2.3 \times 10^7 + 1.79 \times 10^5) \times 5.08 \times 10^4$ | _____ |
| 16 $\frac{6.9 \times 10^8 \times 3.4 \times 10^3}{5.2 \times 10^5}$ | _____ |
| 17 $\frac{8.12 \times 10^{-7}}{3.4 \times 10^5} - 5.79 \times 10^{-3}$ | _____ |

18 $\frac{5.8 \times 10^{10} + 3.42 \times 10^{11}}{7.6 \times 10^4 - 2.9 \times 10^{-3}}$ _____

19 $\frac{6.03 \times 10^{-7} \times 1.98 \times 10^{12}}{7 \times 10^3} + 4.3 \times 10^5$ _____

Key

A	$1.177\,493\,2 \times 10^{12}$
C	8.39×10^6
D	9.9×10^{-3}
E	5 263 158.096
F	15 400 000
G	3.73×10^{-11}
H	3×10^7

I	0.032
K	7.005×10^4
L	0.000 000 507
M	430 170.56
N	5.7899×10^{-3}
O	3.57×10^{-12}

R	4.216×10^{-7}
S	790 800
T	$4.511\,538 \times 10^6$
U	6 005 000 000 000
V	0.000 460 6
W	4.802×10^5